Table 1 Target pairs and experimental activity classes

Target pair	PDB A	PDB B	Dual (n)	A-only (n)	B-only (n)	Neither (n)	Total (n)
EGFR/HER2	3POZ	3RCD	28	37	32	12	109
JAK1/JAK2	6N7A	8BXH	32	32	32	14	110
JAK1/TYK2	6N7A	3LXP	32	32	32	14	110
PIK3CA/mTOR	4L23	4JT6	32	32	32	14	110
AChE/BChE	4EY7	4BDS	28	27	28	15	98
F2/F10	4UDW	2JKH	32	32	32	14	110
PPARG/PPARA	9V8H	6LXA	32	32	32	14	110
PPARA/PPARD	6LXA	5U3Q	32	32	32	14	110

Counts are target-pair–ligand observations; the 867 observations represent 845 distinct ligand entities. Target A and target B follow the order in each pair name. Activity is defined by pChEMBL ≥ 6.0. PDB identifiers refer to the receptors used in the primary analysis.

Table 2 Dual-ligand recovery in matched top-10% candidate lists

Target pair	Scheme	N	k	Dual selected S1 → scheme	Dual recall (%) S1 → scheme
EGFR/HER2	S2	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S3	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S4[a]	97	10	1 → 2	3.57 → 7.14
EGFR/HER2	S5	97	10	1 → 2	3.57 → 7.14
JAK1/JAK2	S2	96	10	6 → 7	18.75 → 21.88
JAK1/JAK2	S3	96	10	6 → 4	18.75 → 12.50
JAK1/JAK2	S4[a]	96	10	6 → 4	18.75 → 12.50
JAK1/JAK2	S5	96	10	6 → 7	18.75 → 21.88
JAK1/TYK2	S2	96	10	3 → 6	9.38 → 18.75
JAK1/TYK2	S3	96	10	3 → 5	9.38 → 15.63
JAK1/TYK2	S4[a]	96	10	3 → 5	9.38 → 15.63
JAK1/TYK2	S5	96	10	3 → 5	9.38 → 15.63
PIK3CA/mTOR	S2	96	10	4 → 6	12.50 → 18.75
PIK3CA/mTOR	S3	96	10	4 → 7	12.50 → 21.88
PIK3CA/mTOR	S4[a]	96	10	4 → 5	12.50 → 15.63
PIK3CA/mTOR	S5	96	10	4 → 7	12.50 → 21.88
AChE/BChE	S2	79	8	6 → 3	22.22 → 11.11
AChE/BChE	S3	80	8	6 → 3	22.22 → 11.11
AChE/BChE	S4[a]	79	8	6 → 3	22.22 → 11.11
AChE/BChE	S5	80	8	6 → 3	22.22 → 11.11
F2/F10	S2	95	10	4 → 4	12.90 → 12.90
F2/F10	S3	96	10	4 → 7	12.50 → 21.88
F2/F10	S4[a]	95	10	4 → 5	12.90 → 16.13
F2/F10	S5	95	10	4 → 6	12.90 → 19.35
PPARG/PPARA	S2	94	10	6 → 2	20.00 → 6.67
PPARG/PPARA	S3	96	10	6 → 4	18.75 → 12.50
PPARG/PPARA	S4[a]	94	10	6 → 3	20.00 → 10.00
PPARG/PPARA	S5	96	10	6 → 1	18.75 → 3.13
PPARA/PPARD	S2	91	10	3 → 5	11.11 → 18.52
PPARA/PPARD	S3	96	10	3 → 6	9.38 → 18.75
PPARA/PPARD	S4[a]	91	10	3 → 4	11.11 → 14.81
PPARA/PPARD	S5	96	10	3 → 5	9.38 → 15.63

Each row uses its own common two-target complete-case population (N); k = ceil(0.10 N). Arrows show S1 → comparison scheme. Dual recall (%) is the fraction of all dual ligands recovered. Counts and recall retain comparison-specific S1 baselines. Full class composition: Table S9.

[a] Supporting comparison against S1.

Table S1 Evaluation-set construction and activity provenance

A. Selection conditions

Target pair	Selection threshold	Chemical selection record	Deposited quota / composition	Eligible n	Selection replay status
EGFR/HER2	6	Murcko cap 5 within class	deposited N=110 with classes 28/38/32/12 before the later unresolved-arm exclusion	109	Documented; not replayed
JAK1/JAK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
JAK1/TYK2	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PIK3CA/mTOR	6	Fixed 110-molecule membership	32 / 32 / 32 / 14	110	Not independently replayed in this compilation
AChE/BChE	6.5 / 5.5	ChEMBL ID-prefix diversity key in deposited script	N=100; dual / A-only / B-only / neither 28/28/28/16	98	Documented; not replayed
F2/F10	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARG/PPARA	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed
PPARA/PPARD	6.5 / 5.5	Small-molecule / size filters; no Murcko cap	dual:32 / A-only: 32 / B-only: 32 / neither: 14	110	Documented; not replayed

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.

Table S1 Evaluation-set construction and activity provenance

B. Activity-source coverage

Target pair	n	Both ends traced	Complete identifiers	Missing identifiers	Source / endpoint mix	Threshold crossing (n)	Value / label mismatch
EGFR/HER2	109	109	101	8	22	13	0
JAK1/JAK2	110	110	109	1	9	0	0
JAK1/TYK2	110	110	108	2	13	4	0
PIK3CA/mTOR	110	110	109	1	24	3	0
AChE/BChE	98	98	98	0	8	2	0
F2/F10	110	110	110	0	10	0	0
PPARG/PPARA	110	110	110	0	42	9	0
PPARA/PPARD	110	110	110	0	40	4	0

All final labels use pChEMBL >= 6.0. Provenance flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied maxima at both endpoints. Across all pairs, cross-document aggregation is present in 81 observations and endpoint heterogeneity in 123 (union 168). The full source crosswalk, raw activity rows, selection evidence and exceptions are included in Supplementary_Data.zip. CAP5 is a post hoc sensitivity analysis and does not redefine initial selection.

Table S2 Receptor structures and docking boxes

A. Receptor identification

Role	Target	PDB	Auth chains	Cognate CCD	Cognate instance	Retained / removed components
Primary	EGFR	3POZ	A	03P	A:1023	Selected protein chain retained
Primary	HER2	3RCD	A	03P	A:9001	Selected protein chain retained
Primary	JAK1	6N7A	AB	KEV	A:1203	Chains A and B retained; PTR converted to TYR
Primary	JAK2	8BXH	A	C87	A:1201	PTR converted to TYR
Primary	TYK2	3LXP	A	IZA	A:1183	PTR converted to TYR
Primary	PIK3CA	4L23	A	X6K	A:1101	p85 removed
Primary	mTOR	4JT6	A	X6K	A:2601	Second mTOR copy and mLST8 removed
Primary	AChE	4EY7	AB	E20	A:604	AChE chains A and B retained
Primary	BChE	4BDS	A	THA	A:701	Selected protein chain retained
Primary	F2	4UDW	HIL	N6L	H:1249	Heavy chain, light chain and hirudin retained
Primary	F10	2JKH	AL	BI7	A:1244	Heavy and light chains retained
Primary	PPARG	9V8H	AB	BRL	A:501	PPARG and PG08-NL peptide retained
Primary	PPARA	6LXA	A	EPA	A:502	Selected protein chain retained
Primary	PPARD	5U3Q	B	7UJ	B:501	PPARD chain B retained
Alternative	AChE	4EY6	AB	GNT	A:604	AChE chains A and B retained
Alternative	BChE	1P0M	A	CHT	A:607	Selected protein chain retained
Alternative	F2	3SHC	HIL	B01	H:3	Heavy chain, light chain and hirudin retained
Alternative	F10	2Y5F	AL	XWG	A:1244	Heavy and light chains retained
Alternative	PPARA	6KAX	A	PLM	A:502	Selected protein chain retained
Alternative	PPARD	5U46	A	7T1	A:505	Selected protein chain retained

Coordinates and dimensions are given in Å. Waters and cocrystal ligands are removed from the receptor. Box centers are bounding-box midpoints; padding is 5 Å with minimum edge 20 Å. Chain identifiers are author chain identifiers. Formal prepared files and preparation exceptions, including observed-fragment templates, are documented in the accompanying source files. Alternative structures cover only the six conditions used in the current analysis; no Q110 alternative-receptor result is included.

Table S2 Receptor structures and docking boxes

B. Docking-box coordinates

Role	Target	PDB	Center (x, y, z), Å	Size (x, y, z), Å
Primary	EGFR	3POZ	18.7460, 31.8320, 11.6260	20.7500, 20.0000, 21.2180
Primary	HER2	3RCD	12.4800, 2.9635, 28.0150	21.1620, 22.1630, 20.0000
Primary	JAK1	6N7A	5.9110, -24.3175, 21.4945	20.0000, 20.5970, 20.0000
Primary	JAK2	8BXH	-12.5897, 14.5939, 21.2626	24.0687, 20.0000, 20.0000
Primary	TYK2	3LXP	-4.4985, 25.5775, -31.0640	20.0000, 20.0000, 20.0000
Primary	PIK3CA	4L23	32.5530, 45.3340, 42.0935	20.0000, 20.0000, 20.0000
Primary	mTOR	4JT6	52.1230, -0.0840, -47.6645	20.4680, 20.0000, 20.0000
Primary	AChE	4EY7	-13.9875, -43.9055, 27.1085	23.3410, 20.0000, 20.3550
Primary	BChE	4BDS	133.0760, 116.1130, 41.3350	20.0000, 20.0000, 20.0000
Primary	F2	4UDW	129.7390, -14.4025, 164.9460	20.6000, 20.0000, 20.0000
Primary	F10	2JKH	-3.6430, 10.7695, 22.7710	20.0000, 20.0000, 21.1180
Primary	PPARG	9V8H	-5.4540, -5.3730, -23.5915	20.0000, 20.1400, 20.0000
Primary	PPARA	6LXA	11.9975, 5.7420, -7.4430	20.0000, 20.2000, 23.4320
Primary	PPARD	5U3Q	45.4160, -37.6795, 103.9765	20.0000, 20.0000, 22.4490
Alternative	AChE	4EY6	-10.4920, -43.5255, 29.4535	20, 20, 20
Alternative	BChE	1P0M	133.5840, 116.7965, 40.3950	20, 20, 20
Alternative	F2	3SHC	16.7620, -12.0230, 22.9065	20.5680, 20, 20
Alternative	F10	2Y5F	-4.6240, 10.6510, 22.4765	20, 20.2300, 21.4290
Alternative	PPARA	6KAX	12.0005, 6.0880, -7.6585	20, 20, 22.6890
Alternative	PPARD	5U46	45.4215, -1.7525, 135.5685	20, 20, 20.4870

Coordinates and dimensions are given in Å. Waters and cocrystal ligands are removed from the receptor. Box centers are bounding-box midpoints; padding is 5 Å with minimum edge 20 Å. Chain identifiers are author chain identifiers. Formal prepared files and preparation exceptions, including observed-fragment templates, are documented in the accompanying source files. Alternative structures cover only the six conditions used in the current analysis; no Q110 alternative-receptor result is included.

Table S3 Cocrystal-ligand redocking across five random seeds

Target	PDB	CCD	Seeds	Jobs	Top-1 RMSD median [min, max] (Å)	Best-saved RMSD median [min, max] (Å)	Top-1 <2 Å (n)	Best <2 Å (n)	Modes
EGFR	3POZ	03P	5	5	1.0960 [1.0940, 1.1170]	1.0960 [1.0940, 1.1170]	5	5	9
HER2	3RCD	03P	5	5	2.0540 [2.0210, 2.8800]	2.0460 [1.3370, 2.0580]	0	1	9
JAK1	6N7A	KEV	5	5	0.4560 [0.4440, 0.4740]	0.4560 [0.4440, 0.4740]	5	5	9
JAK2	8BXH	C87	5	5	10.6940 [10.6860, 10.7200]	0.9980 [0.9490, 1.0750]	0	5	9
TYK2	3LXP	IZA	5	5	0.2560 [0.2180, 0.2620]	0.2560 [0.2180, 0.2620]	5	5	9
PIK3CA	4L23	X6K	5	5	0.9680 [0.8960, 0.9790]	0.9680 [0.8960, 0.9790]	5	5	9
mTOR	4JT6	X6K	5	5	4.9280 [4.9270, 4.9330]	0.5810 [0.5550, 0.6010]	0	5	9
AChE	4EY7	E20	5	5	0.8030 [0.7870, 0.8280]	0.8030 [0.7870, 0.8280]	5	5	9
BChE	4BDS	THA	5	5	4.7540 [4.7530, 4.7540]	0.3140 [0.3040, 0.4290]	0	5	9
F2	4UDW	N6L	5	5	0.5380 [0.5270, 0.5680]	0.5380 [0.5270, 0.5680]	5	5	9
F10	2JKH	BI7	5	5	0.8220 [0.7790, 0.8480]	0.8220 [0.7790, 0.8480]	5	5	9
PPARG	9V8H	BRL	5	5	2.0810 [1.6560, 6.7880]	1.4400 [0.7280, 1.9770]	2	5	9
PPARA	6LXA	EPA	5	5	7.7870 [7.7100, 7.8410]	1.1250 [0.5970, 1.6120]	0	5	9
PPARD	5U3Q	7UJ	5	5	0.4840 [0.4460, 0.7780]	0.4840 [0.4460, 0.7780]	5	5	9

Seeds: 17, 29, 42, 71 and 101. Symmetry-aware heavy-atom RMSD is calculated in the receptor coordinate frame. Top-ranked-pose success and success among saved poses are separate outcomes. The summary uses recorded RMSDs without recomputing coordinates; every seed-level record is in Supplementary_Data.zip. CCD denotes the cognate ligand Chemical Component Dictionary identifier. The "Jobs" and "Modes" columns report the number of successful docking jobs and the number of saved poses per job, respectively.

Table S4 Analysis populations and score availability

A. Directional and chemistry populations

Target pair	Direction	Labeled n	S1 directional n	S1 missing n	Unassigned fold after S1	Chemistry OOF n
EGFR/HER2	Dual vs A-only	65	65	0	0	65
EGFR/HER2	Dual vs B-only	60	60	0	1	59
JAK1/JAK2	Dual vs A-only	64	64	0	0	64
JAK1/JAK2	Dual vs B-only	64	64	0	0	64
JAK1/TYK2	Dual vs A-only	64	64	0	1	63
JAK1/TYK2	Dual vs B-only	64	64	0	1	63
PIK3CA/mTOR	Dual vs A-only	64	64	0	0	64
PIK3CA/mTOR	Dual vs B-only	64	64	0	0	64
AChE/BChE	Dual vs A-only	55	52	3	0	52
AChE/BChE	Dual vs B-only	56	55	1	0	55
F2/F10	Dual vs A-only	64	64	0	1	63
F2/F10	Dual vs B-only	64	64	0	1	63
PPARG/PPARA	Dual vs A-only	64	64	0	1	63
PPARG/PPARA	Dual vs B-only	64	64	0	0	64
PPARA/PPARD	Dual vs A-only	64	64	0	0	64
PPARA/PPARD	Dual vs B-only	64	64	0	0	64

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.

Table S4 Analysis populations and score availability

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
EGFR/HER2	S1	109	109	0	109	109	0
EGFR/HER2	S2	109	109	0	109	109	0
EGFR/HER2	S3	109	109	0	109	109	0
EGFR/HER2	S4	109	109	0	109	109	0
EGFR/HER2	S5	109	109	0	109	109	0
JAK1/JAK2	S1	110	110	0	110	110	0
JAK1/JAK2	S2	110	110	0	110	110	0
JAK1/JAK2	S3	110	110	0	110	110	0
JAK1/JAK2	S4	110	110	0	110	110	0
JAK1/JAK2	S5	110	110	0	110	110	0
JAK1/TYK2	S1	110	110	0	110	110	0
JAK1/TYK2	S2	110	110	0	110	110	0
JAK1/TYK2	S3	110	110	0	110	110	0
JAK1/TYK2	S4	110	110	0	110	110	0
JAK1/TYK2	S5	110	110	0	110	110	0
PIK3CA/mTOR	S1	110	110	0	110	110	0
PIK3CA/mTOR	S2	110	110	0	110	110	0
PIK3CA/mTOR	S3	110	110	0	110	110	0
PIK3CA/mTOR	S4	110	110	0	110	110	0
PIK3CA/mTOR	S5	110	110	0	110	110	0
AChE/BChE	S1	98	95	3	98	95	3
AChE/BChE	S2	98	94	4	98	94	4
AChE/BChE	S3	98	95	3	98	95	3
AChE/BChE	S4	98	94	4	98	94	4
AChE/BChE	S5	98	97	1	98	97	1
F2/F10	S1	110	110	0	110	110	0
F2/F10	S2	110	109	1	110	110	0
F2/F10	S3	110	110	0	110	110	0
F2/F10	S4	110	109	1	110	110	0
F2/F10	S5	110	108	2	110	108	2

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.

Table S4 Analysis populations and score availability (continued)

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
PPARG/PPARA	S1	110	110	0	110	110	0
PPARG/PPARA	S2	110	108	2	110	108	2
PPARG/PPARA	S3	110	110	0	110	110	0
PPARG/PPARA	S4	110	108	2	110	108	2
PPARG/PPARA	S5	110	110	0	110	110	0
PPARA/PPARD	S1	110	110	0	110	110	0
PPARA/PPARD	S2	110	105	5	110	105	5
PPARA/PPARD	S3	110	110	0	110	110	0
PPARA/PPARD	S4	110	105	5	110	105	5
PPARA/PPARD	S5	110	110	0	110	110	0

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.

Table S5 Negative-class and pocket-correspondence controls

A. Negative-class replacement

Target pair	Direction	n (dual / single / neither)	AUROC single-target	AUROC neither	Difference (95% CI)
EGFR/HER2	Dual vs A-only	28 / 37 / 12	0.6274	0.7083	0.0809 [-0.1447, 0.2871]
EGFR/HER2	Dual vs B-only	28 / 32 / 12	0.2467	0.7470	0.5004 [0.3025, 0.6819]
JAK1/JAK2	Dual vs A-only	32 / 32 / 14	0.5908	0.7344	0.1436 [-0.0498, 0.3326]
JAK1/JAK2	Dual vs B-only	32 / 32 / 14	0.7485	0.7143	-0.0342 [-0.2238, 0.1410]
JAK1/TYK2	Dual vs A-only	32 / 32 / 14	0.6475	0.7321	0.0847 [-0.0979, 0.2578]
JAK1/TYK2	Dual vs B-only	32 / 32 / 14	0.4219	0.8058	0.3839 [0.1953, 0.5693]
PIK3CA/mTOR	Dual vs A-only	32 / 32 / 14	0.7363	0.7679	0.0315 [-0.0897, 0.1497]
PIK3CA/mTOR	Dual vs B-only	32 / 32 / 14	0.5801	0.6964	0.1164 [-0.0592, 0.2872]
AChE/BChE	Dual vs A-only	27 / 25 / 15	0.6341	0.6963	0.0622 [-0.1047, 0.2242]
AChE/BChE	Dual vs B-only	27 / 28 / 15	0.6442	0.5679	-0.0763 [-0.2019, 0.0469]
F2/F10	Dual vs A-only	32 / 32 / 14	0.4082	0.5045	0.0963 [-0.0466, 0.2538]
F2/F10	Dual vs B-only	32 / 32 / 14	0.3623	0.4888	0.1265 [-0.0172, 0.2755]
PPARG/PPARA	Dual vs A-only	32 / 32 / 14	0.6553	0.7567	0.1014 [-0.0724, 0.2727]
PPARG/PPARA	Dual vs B-only	32 / 32 / 14	0.6021	0.7054	0.1033 [-0.1187, 0.3121]
PPARA/PPARD	Dual vs A-only	32 / 32 / 14	0.6787	0.6518	-0.0269 [-0.2102, 0.1486]
PPARA/PPARD	Dual vs B-only	32 / 32 / 14	0.5107	0.4866	-0.0241 [-0.1915, 0.1482]

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.

Table S5 Negative-class and pocket-correspondence controls

B. Corresponding and noncorresponding pockets

Target pair	Direction	n (dual / negative)	Corresp. AUROC	Noncorresp. AUROC	Difference (95% CI)	Additional exclusions
EGFR/HER2	Dual vs A-only	28 / 37	0.6274	0.6573	-0.0299 [-0.0936, 0.0271]	0
EGFR/HER2	Dual vs B-only	28 / 32	0.2467	0.2355	0.0112 [-0.0681, 0.0949]	0
JAK1/JAK2	Dual vs A-only	32 / 32	0.5908	0.5908	0 [-0.0771, 0.0703]	0
JAK1/JAK2	Dual vs B-only	32 / 32	0.7485	0.7012	0.0474 [-0.0278, 0.1338]	0
JAK1/TYK2	Dual vs A-only	32 / 32	0.6475	0.6250	0.0225 [-0.0898, 0.1397]	0
JAK1/TYK2	Dual vs B-only	32 / 32	0.4219	0.4634	-0.0415 [-0.1279, 0.0410]	0
PIK3CA/mTOR	Dual vs A-only	32 / 32	0.7363	0.6519	0.0845 [-0.0454, 0.2144]	0
PIK3CA/mTOR	Dual vs B-only	32 / 32	0.5801	0.5859	-0.0059 [-0.1230, 0.1094]	0
AChE/BChE	Dual vs A-only	27 / 25	0.6341	0.4681	0.1659 [0.0548, 0.2830]	0
AChE/BChE	Dual vs B-only	27 / 28	0.6442	0.5807	0.0635 [-0.0661, 0.1971]	0
F2/F10	Dual vs A-only	32 / 32	0.4082	0.3926	0.0156 [-0.0957, 0.1230]	0
F2/F10	Dual vs B-only	32 / 32	0.3623	0.3887	-0.0264 [-0.1465, 0.0967]	0
PPARG/PPARA	Dual vs A-only	32 / 32	0.6553	0.5928	0.0625 [-0.1035, 0.2275]	0
PPARG/PPARA	Dual vs B-only	32 / 32	0.6021	0.6909	-0.0889 [-0.2563, 0.0869]	0
PPARA/PPARD	Dual vs A-only	32 / 32	0.6787	0.4434	0.2354 [0.0693, 0.4043]	0
PPARA/PPARD	Dual vs B-only	32 / 32	0.5107	0.4561	0.0547 [-0.1436, 0.2441]	0

Panel A difference = neither minus single-target negative class. Panel B difference = corresponding minus noncorresponding pocket. All estimates use S1. Individual AUROC confidence intervals and ligand identifiers are retained in the complete source CSVs. Corresp., corresponding pocket; noncorresp., noncorresponding pocket. Grouped counts follow the class order in the column heading.

Table S6 Out-of-fold performance of ligand-chemistry models

Target pair	Direction	Common OOF n	Descriptor AUROC	ECFP4 AUROC	ECFP4 + S1 AUROC	Matched S1 AUROC	ECFP4 + S1 increment
EGFR/HER2	Dual vs A-only	65	0.6757	0.8002	0.7838	0.6274	-0.0164
EGFR/HER2	Dual vs B-only	59	0.7592	0.8790	0.8767	0.2408	-0.0023
JAK1/JAK2	Dual vs A-only	64	0.6030	0.9185	0.9199	0.5908	0.0015
JAK1/JAK2	Dual vs B-only	64	0.9014	0.9678	0.9707	0.7485	0.0029
JAK1/TYK2	Dual vs A-only	63	0.5655	0.8488	0.8438	0.6371	-0.0050
JAK1/TYK2	Dual vs B-only	63	0.8407	0.9042	0.9032	0.4032	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	0.6094	0.8213	0.8428	0.7363	0.0215
PIK3CA/mTOR	Dual vs B-only	64	0.2939	0.7285	0.7256	0.5801	-0.0029
AChE/BChE	Dual vs A-only	52	0.6815	0.8993	0.8948	0.6341	-0.0044
AChE/BChE	Dual vs B-only	55	0.7884	0.8201	0.8148	0.6442	-0.0053
F2/F10	Dual vs A-only	63	0.6139	0.9405	0.9335	0.4204	-0.0071
F2/F10	Dual vs B-only	63	0.4889	0.6633	0.6996	0.3740	0.0363
PPARG/PPARA	Dual vs A-only	63	0.7198	0.8206	0.8065	0.6764	-0.0141
PPARG/PPARA	Dual vs B-only	64	0.7158	0.6670	0.6611	0.6021	-0.0059
PPARA/PPARD	Dual vs A-only	64	0.4375	0.9316	0.9287	0.6787	-0.0029
PPARA/PPARD	Dual vs B-only	64	0.3115	0.8740	0.8789	0.5107	0.0049

All models and S1 are evaluated on identical members within each direction. The increment is AUROC(ECFP4 + S1) minus AUROC(ECFP4). Existing frozen five-fold assignments are retained; Q110 uses sorted canonical Murcko scaffolds assigned cyclically to folds 0-4. Standardization is fitted in training folds. Full descriptors, scaffolds, similarities, folds and OOF predictions are supplied as data files; no new model is fitted during table preparation. OOF refers to the learned chemistry models; S1 is a direct docking score.

Table S7 Random-seed and receptor-structure sensitivity

A. Five-seed distributions

Target pair	Metric	Seed 42	Median	Minimum	Maximum	IQR	Finite seeds
EGFR/HER2	AUROC B	0.6274	0.6375	0.6274	0.6477	0.0111	5
EGFR/HER2	AUROC A	0.2467	0.2388	0.2305	0.2467	0.0089	5
EGFR/HER2	Minimum AUROC	0.2467	0.2388	0.2305	0.2467	0.0089	5
JAK1/JAK2	AUROC B	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/JAK2	AUROC A	0.7485	0.7485	0.7422	0.7573	0.0112	5
JAK1/JAK2	Minimum AUROC	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/TYK2	AUROC B	0.6475	0.6475	0.6328	0.6631	0.0156	5
JAK1/TYK2	AUROC A	0.4219	0.4229	0.4121	0.4302	0.0015	5
JAK1/TYK2	Minimum AUROC	0.4219	0.4229	0.4121	0.4302	0.0015	5
PIK3CA/mTOR	AUROC B	0.7363	0.7354	0.6997	0.7402	0.0029	5
PIK3CA/mTOR	AUROC A	0.5801	0.5752	0.5596	0.5869	0.0190	5
PIK3CA/mTOR	Minimum AUROC	0.5801	0.5752	0.5596	0.5869	0.0190	5
AChE/BChE	AUROC B	0.6341	0.6422	0.6049	0.6533	0.0163	5
AChE/BChE	AUROC A	0.6442	0.6442	0.6356	0.6448	0.0026	5
AChE/BChE	Minimum AUROC	0.6341	0.6356	0.6049	0.6448	0.0075	5
F2/F10	AUROC B	0.4082	0.4062	0.3955	0.4082	0.0015	5
F2/F10	AUROC A	0.3623	0.3647	0.3594	0.3682	0.0029	5
F2/F10	Minimum AUROC	0.3623	0.3647	0.3594	0.3682	0.0029	5
PPARG/PPARA	AUROC B	0.6553	0.6553	0.6465	0.6680	0.0127	5
PPARG/PPARA	AUROC A	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARG/PPARA	Minimum AUROC	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARA/PPARD	AUROC B	0.6787	0.6929	0.6787	0.6953	0.0039	5
PPARA/PPARD	AUROC A	0.5107	0.5088	0.4800	0.5151	0.0146	5
PPARA/PPARD	Minimum AUROC	0.5107	0.5088	0.4800	0.5151	0.0146	5

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.

Table S7 Random-seed and receptor-structure sensitivity

B. One-receptor substitutions

Target pair	Alt. PDB	Side	n (dual / A-only / B-only)	Original minimum AUROC	Alternative minimum AUROC	Minimum AUROC difference (95% CI)	Affected AUROC difference	Unreplaced AUROC difference
AChE/BChE	1P0M	B	26 / 23 / 26	0.6037	0.6003	-0.0033 [-0.0803, 0.0836]	-0.0033	0
AChE/BChE	4EY6	A	27 / 23 / 26	0.6184	0.5670	-0.0514 [-0.1852, 0]	-0.0940	0
F2/F10	2Y5F	B	30 / 32 / 31	0.3828	0.3828	0 [-0.0469, 0.0229]	-0.0146	0
F2/F10	3SHC	A	30 / 32 / 31	0.3828	0.4333	0.0505 [0, 0.1441]	0.0780	0
PPARA/PPARD	5U46	B	32 / 32 / 32	0.5107	0.5107	0 [-0.0225, 0]	-0.0181	0
PPARA/PPARD	6KAX	A	32 / 32 / 32	0.5107	0.4438	-0.0669 [-0.1484, 0.0044]	-0.0669	0

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.

Table S8 Chemical-diversity and activity-threshold sensitivity

A. CAP5 membership

Target pair	Original n	CAP5 n	Excluded n	Excluded dual	Excluded A-only	Excluded B-only	Excluded neither
EGFR/HER2	109	109	0	0	0	0	0
JAK1/JAK2	110	107	3	0	3	0	0
JAK1/TYK2	110	110	0	0	0	0	0
PIK3CA/mTOR	110	110	0	0	0	0	0
AChE/BChE	98	98	0	0	0	0	0
F2/F10	110	110	0	0	0	0	0
PPARG/PPARA	110	110	0	0	0	0	0
PPARA/PPARD	110	107	3	0	1	2	0

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

B. Vina minimum AUROC before and after CAP5

Target pair	Primary minimum AUROC (95% CI)	CAP5 minimum AUROC (95% CI)	Point change
EGFR/HER2	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]	0.0000
JAK1/JAK2	0.5908 [0.4512, 0.7246]	0.6315 [0.4871, 0.7597]	0.0406
JAK1/TYK2	0.4219 [0.2812, 0.5635]	0.4219 [0.2812, 0.5635]	0.0000
PIK3CA/mTOR	0.5801 [0.4346, 0.7129]	0.5801 [0.4346, 0.7129]	0.0000
AChE/BChE	0.6341 [0.4607, 0.7540]	0.6341 [0.4607, 0.7540]	0.0000
F2/F10	0.3623 [0.2305, 0.4805]	0.3623 [0.2305, 0.4805]	0.0000
PPARG/PPARA	0.6021 [0.4561, 0.7051]	0.6021 [0.4561, 0.7051]	0.0000
PPARA/PPARD	0.5107 [0.3652, 0.6475]	0.4969 [0.3427, 0.6354]	-0.0139

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

C. PIK3CA/mTOR strict 6.5/5.5 subset

Target pair	Scheme	n (dual / A-only / B-only)	AUROC, target B (95% CI)	AUROC, target A (95% CI)	Minimum AUROC (95% CI)
PIK3CA/mTOR	S1	25 / 12 / 14	0.7900 [0.6200, 0.9300]	0.4571 [0.2743, 0.6429]	0.4571 [0.2743, 0.6429]
PIK3CA/mTOR	S2	25 / 12 / 14	0.6200 [0.4133, 0.8133]	0.5343 [0.3429, 0.7229]	0.5343 [0.3343, 0.6829]
PIK3CA/mTOR	S3	25 / 12 / 14	0.7500 [0.5767, 0.8933]	0.6286 [0.4457, 0.7971]	0.6286 [0.4429, 0.7800]
PIK3CA/mTOR	S4	25 / 12 / 14	0.6267 [0.4267, 0.8167]	0.6057 [0.4143, 0.7829]	0.6057 [0.3867, 0.7229]
PIK3CA/mTOR	S5	25 / 12 / 14	0.5933 [0.4033, 0.7833]	0.6686 [0.4828, 0.8371]	0.5933 [0.4000, 0.7367]

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S8 Chemical-diversity and activity-threshold sensitivity

D. Chemistry-model sensitivity to CAP5

Target pair	Direction	Original OOF n	CAP5 OOF n	Original ECFP4 AUROC	CAP5 ECFP4 AUROC	ECFP4 AUROC change	Original Vina increment	CAP5 Vina increment
EGFR/HER2	Dual vs A-only	65	65	0.8002	0.8002	0.0000	-0.0164	-0.0164
EGFR/HER2	Dual vs B-only	59	59	0.8790	0.8790	0.0000	-0.0023	-0.0023
JAK1/JAK2	Dual vs A-only	64	61	0.9185	0.9079	-0.0106	0.0015	0.0048
JAK1/JAK2	Dual vs B-only	64	64	0.9678	0.9678	0.0000	0.0029	0.0029
JAK1/TYK2	Dual vs A-only	63	63	0.8488	0.8488	0.0000	-0.0050	-0.0050
JAK1/TYK2	Dual vs B-only	63	63	0.9042	0.9042	0.0000	-0.0010	-0.0010
PIK3CA/mTOR	Dual vs A-only	64	64	0.8213	0.8213	0.0000	0.0215	0.0215
PIK3CA/mTOR	Dual vs B-only	64	64	0.7285	0.7285	0.0000	-0.0029	-0.0029
AChE/BChE	Dual vs A-only	52	52	0.8993	0.8993	0.0000	-0.0044	-0.0044
AChE/BChE	Dual vs B-only	55	55	0.8201	0.8201	0.0000	-0.0053	-0.0053
F2/F10	Dual vs A-only	63	63	0.9405	0.9405	0.0000	-0.0071	-0.0071
F2/F10	Dual vs B-only	63	63	0.6633	0.6633	0.0000	0.0363	0.0363
PPARG/PPARA	Dual vs A-only	63	63	0.8206	0.8206	0.0000	-0.0141	-0.0141
PPARG/PPARA	Dual vs B-only	64	64	0.6670	0.6670	0.0000	-0.0059	-0.0059
PPARA/PPARD	Dual vs A-only	64	63	0.9316	0.9345	0.0028	-0.0029	-0.0030
PPARA/PPARD	Dual vs B-only	64	62	0.8740	0.8688	-0.0053	0.0049	0.0042

CAP5 is a post hoc sensitivity analysis: at most five ligands per nonempty Murcko scaffold within pair and activity class; acyclic ligands are exempt. Six ligands are excluded without replacement. Complete CAP5 method comparisons, chemical models, controls and ranking outputs are retained in Supplementary_Data.zip. The PPARA/PPARD S5 dual-versus-A-only CI changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]; this boundary change is retained, without a general significance claim. The strict Q110 subset has 56 members (25 dual, 12 A-only, 14 B-only, 5 neither). Main and subset populations differ; no new paired CI for their difference is calculated.

Table S9 Joint-ranking composition and directional concordance

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
EGFR/HER2	S1	97	10	28	1	4	5
EGFR/HER2	S2	97	10	28	2	1	7
EGFR/HER2	S3	97	10	28	2	4	4
EGFR/HER2	S4	97	10	28	2	2	6
EGFR/HER2	S5	97	10	28	2	0	8
JAK1/JAK2	S1	96	10	32	6	3	1
JAK1/JAK2	S2	96	10	32	7	3	0
JAK1/JAK2	S3	96	10	32	4	4	2
JAK1/JAK2	S4	96	10	32	4	3	3
JAK1/JAK2	S5	96	10	32	7	3	0
JAK1/TYK2	S1	96	10	32	3	1	6
JAK1/TYK2	S2	96	10	32	6	2	2
JAK1/TYK2	S3	96	10	32	5	0	5
JAK1/TYK2	S4	96	10	32	5	3	2
JAK1/TYK2	S5	96	10	32	5	4	1
PIK3CA/mTOR	S1	96	10	32	4	0	6
PIK3CA/mTOR	S2	96	10	32	6	3	1
PIK3CA/mTOR	S3	96	10	32	7	1	2
PIK3CA/mTOR	S4	96	10	32	5	3	2
PIK3CA/mTOR	S5	96	10	32	7	1	2
AChE/BChE	S1	80	8	27	6	0	2
AChE/BChE	S2	79	8	27	3	2	3
AChE/BChE	S3	80	8	27	3	3	2
AChE/BChE	S4	79	8	27	3	3	2
AChE/BChE	S5	82	9	27	3	1	5
F2/F10	S1	96	10	32	4	4	2
F2/F10	S2	95	10	31	4	3	3
F2/F10	S3	96	10	32	7	1	2
F2/F10	S4	95	10	31	5	5	0
F2/F10	S5	95	10	31	6	3	1

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

A1. Method-specific top-10% counts

Target pair	Scheme	N	k	Dual in population	Dual selected	A-only selected	B-only selected
PPARG/PPARA	S1	96	10	32	6	3	1
PPARG/PPARA	S2	94	10	30	2	4	4
PPARG/PPARA	S3	96	10	32	4	3	3
PPARG/PPARA	S4	94	10	30	3	4	3
PPARG/PPARA	S5	96	10	32	1	4	5
PPARA/PPARD	S1	96	10	32	3	2	5
PPARA/PPARD	S2	91	10	27	5	2	3
PPARA/PPARD	S3	96	10	32	6	2	2
PPARA/PPARD	S4	91	10	27	4	1	5
PPARA/PPARD	S5	96	10	32	5	1	4

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
EGFR/HER2	S1	0.1000	0.9000	0.0357	0.2887
EGFR/HER2	S2	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S3	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S4	0.2000	0.8000	0.0714	0.2887
EGFR/HER2	S5	0.2000	0.8000	0.0714	0.2887
JAK1/JAK2	S1	0.6000	0.4000	0.1875	0.3333
JAK1/JAK2	S2	0.7000	0.3000	0.2188	0.3333
JAK1/JAK2	S3	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S4	0.4000	0.6000	0.1250	0.3333
JAK1/JAK2	S5	0.7000	0.3000	0.2188	0.3333
JAK1/TYK2	S1	0.3000	0.7000	0.0938	0.3333
JAK1/TYK2	S2	0.6000	0.4000	0.1875	0.3333
JAK1/TYK2	S3	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S4	0.5000	0.5000	0.1562	0.3333
JAK1/TYK2	S5	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S1	0.4000	0.6000	0.1250	0.3333
PIK3CA/mTOR	S2	0.6000	0.4000	0.1875	0.3333
PIK3CA/mTOR	S3	0.7000	0.3000	0.2188	0.3333
PIK3CA/mTOR	S4	0.5000	0.5000	0.1562	0.3333
PIK3CA/mTOR	S5	0.7000	0.3000	0.2188	0.3333
AChE/BChE	S1	0.7500	0.2500	0.2222	0.3375
AChE/BChE	S2	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S3	0.3750	0.6250	0.1111	0.3375
AChE/BChE	S4	0.3750	0.6250	0.1111	0.3418
AChE/BChE	S5	0.3333	0.6667	0.1111	0.3293
F2/F10	S1	0.4000	0.6000	0.1250	0.3333
F2/F10	S2	0.4000	0.6000	0.1290	0.3263
F2/F10	S3	0.7000	0.3000	0.2188	0.3333
F2/F10	S4	0.5000	0.5000	0.1613	0.3263
F2/F10	S5	0.6000	0.4000	0.1935	0.3263

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

A2. Method-specific top-10% fractions

Target pair	Scheme	Dual fraction	Single-target fraction	Dual recall	Population dual fraction
PPARG/PPARA	S1	0.6000	0.4000	0.1875	0.3333
PPARG/PPARA	S2	0.2000	0.8000	0.0667	0.3191
PPARG/PPARA	S3	0.4000	0.6000	0.1250	0.3333
PPARG/PPARA	S4	0.3000	0.7000	0.1000	0.3191
PPARG/PPARA	S5	0.1000	0.9000	0.0312	0.3333
PPARA/PPARD	S1	0.3000	0.7000	0.0938	0.3333
PPARA/PPARD	S2	0.5000	0.5000	0.1852	0.2967
PPARA/PPARD	S3	0.6000	0.4000	0.1875	0.3333
PPARA/PPARD	S4	0.4000	0.6000	0.1481	0.2967
PPARA/PPARD	S5	0.5000	0.5000	0.1562	0.3333

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
EGFR/HER2	S2	97	28	-0.0257	-0.1000	Discordant	NO
EGFR/HER2	S3	97	28	0.0167	-0.1000	Concordant improvement	NO
EGFR/HER2	S4	97	28	0.0346	-0.1000	Concordant improvement	YES
EGFR/HER2	S5	97	28	-0.0011	-0.1000	Discordant	NO
JAK1/JAK2	S2	96	32	-0.1523	-0.1000	Discordant	NO
JAK1/JAK2	S3	96	32	-0.1172	0.2000	Concordant worsening	NO
JAK1/JAK2	S4	96	32	-0.1465	0.2000	Concordant worsening	YES
JAK1/JAK2	S5	96	32	-0.0771	-0.1000	Discordant	NO
JAK1/TYK2	S2	96	32	0.1377	-0.3000	Concordant improvement	NO
JAK1/TYK2	S3	96	32	0.0654	-0.2000	Concordant improvement	NO
JAK1/TYK2	S4	96	32	0.2539	-0.2000	Concordant improvement	YES
JAK1/TYK2	S5	96	32	0.2471	-0.2000	Concordant improvement	NO
PIK3CA/mTOR	S2	96	32	-0.0049	-0.2000	Discordant	NO
PIK3CA/mTOR	S3	96	32	0.1240	-0.3000	Concordant improvement	NO
PIK3CA/mTOR	S4	96	32	-0.0635	-0.1000	Discordant	YES
PIK3CA/mTOR	S5	96	32	0.0176	-0.3000	Concordant improvement	NO
AChE/BChE	S2	79	27	-0.1416	0.3750	Concordant worsening	NO
AChE/BChE	S3	80	27	-0.1433	0.3750	Concordant worsening	NO
AChE/BChE	S4	79	27	-0.2431	0.3750	Concordant worsening	YES
AChE/BChE	S5	80	27	-0.1830	0.3750	Concordant worsening	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance (continued)

B. Paired changes versus Vina

Target pair	Scheme vs S1	Ranking common n	Directional common dual n	Minimum AUROC difference	Single-target fraction difference	Concordance	Supporting comparison
F2/F10	S2	95	31	0.1643	0	At least one metric unchanged	NO
F2/F10	S3	96	32	0.2471	-0.3000	Concordant improvement	NO
F2/F10	S4	95	31	0.1956	-0.1000	Concordant improvement	YES
F2/F10	S5	95	31	0.1815	-0.2000	Concordant improvement	NO
PPARG/PPARA	S2	94	30	-0.3089	0.4000	Concordant worsening	NO
PPARG/PPARA	S3	96	32	-0.2915	0.2000	Concordant worsening	NO
PPARG/PPARA	S4	94	30	-0.2578	0.3000	Concordant worsening	YES
PPARG/PPARA	S5	96	32	-0.1558	0.5000	Concordant worsening	NO
PPARA/PPARD	S2	91	27	0.1192	-0.2000	Concordant improvement	NO
PPARA/PPARD	S3	96	32	0.1582	-0.3000	Concordant improvement	NO
PPARA/PPARD	S4	91	27	0	-0.1000	At least one metric unchanged	YES
PPARA/PPARD	S5	96	32	0.1230	-0.2000	Concordant improvement	NO

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S9 Joint-ranking composition and directional concordance

C. Concordance counts

Scheme vs S1	Pairs	Concordant improvement	Concordant worsening	Discordant	At least one metric unchanged
S2	8	2	2	3	1
S3	8	5	3	0	0
S4	8	3	3	1	1
S5	8	4	2	2	0

Panel A uses each method's own members and is descriptive. Table 2 and panel B use separate matched populations for every comparison. Joint rank = max(rank A, rank B), with average single-target ranks; ties are resolved by rank sum then global molecular ID. Both differences are new scheme minus S1. Opposite signs define concordant improvement or worsening; equal nonzero signs are discordant; a zero value for either difference indicates that at least one metric is unchanged. S4-S1 is supporting, using verified identical directional members and no additional CI. Complete paired composition, per-ligand ranks and all CAP5 ranking results are supplied as data files.

Table S10 Directional discrimination and paired scoring-workflow comparisons

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
EGFR/HER2	S1	28 / 37 / 32	0.6274 [0.4865, 0.7645]	0.2467 [0.1328, 0.3783]	0.2467 [0.1328, 0.3783]
EGFR/HER2	S2	28 / 37 / 32	0.3731 [0.2355, 0.5150]	0.2210 [0.0993, 0.3538]	0.2210 [0.0993, 0.3516]
EGFR/HER2	S3	28 / 37 / 32	0.5241 [0.3793, 0.6641]	0.2634 [0.1339, 0.4051]	0.2634 [0.1339, 0.4051]
EGFR/HER2	S4	28 / 37 / 32	0.4204 [0.2760, 0.5690]	0.2812 [0.1551, 0.4208]	0.2812 [0.1551, 0.4163]
EGFR/HER2	S5	28 / 37 / 32	0.4344 [0.2896, 0.5820]	0.2455 [0.1239, 0.3806]	0.2455 [0.1239, 0.3806]
JAK1/JAK2	S1	32 / 32 / 32	0.5908 [0.4512, 0.7275]	0.7485 [0.6235, 0.8604]	0.5908 [0.4512, 0.7246]
JAK1/JAK2	S2	32 / 32 / 32	0.4385 [0.2979, 0.5791]	0.7402 [0.6055, 0.8623]	0.4385 [0.2979, 0.5791]
JAK1/JAK2	S3	32 / 32 / 32	0.4736 [0.3291, 0.6211]	0.7480 [0.6162, 0.8623]	0.4736 [0.3291, 0.6201]
JAK1/JAK2	S4	32 / 32 / 32	0.4443 [0.2988, 0.5898]	0.6846 [0.5430, 0.8184]	0.4443 [0.2988, 0.5898]
JAK1/JAK2	S5	32 / 32 / 32	0.5137 [0.3682, 0.6592]	0.7012 [0.5605, 0.8311]	0.5137 [0.3682, 0.6592]
JAK1/TYK2	S1	32 / 32 / 32	0.6475 [0.5049, 0.7812]	0.4219 [0.2812, 0.5654]	0.4219 [0.2812, 0.5635]
JAK1/TYK2	S2	32 / 32 / 32	0.6631 [0.5225, 0.7940]	0.5596 [0.4102, 0.7021]	0.5596 [0.4102, 0.6836]
JAK1/TYK2	S3	32 / 32 / 32	0.7090 [0.5742, 0.8271]	0.4873 [0.3379, 0.6338]	0.4873 [0.3379, 0.6309]
JAK1/TYK2	S4	32 / 32 / 32	0.7637 [0.6348, 0.8779]	0.6758 [0.5254, 0.8164]	0.6758 [0.5254, 0.7998]
JAK1/TYK2	S5	32 / 32 / 32	0.6689 [0.5322, 0.8067]	0.6846 [0.5430, 0.8115]	0.6689 [0.5176, 0.7666]
PIK3CA/mTOR	S1	32 / 32 / 32	0.7363 [0.6074, 0.8486]	0.5801 [0.4346, 0.7188]	0.5801 [0.4346, 0.7129]
PIK3CA/mTOR	S2	32 / 32 / 32	0.5752 [0.4297, 0.7168]	0.6592 [0.5195, 0.7871]	0.5752 [0.4287, 0.6982]
PIK3CA/mTOR	S3	32 / 32 / 32	0.7041 [0.5674, 0.8281]	0.7305 [0.5986, 0.8467]	0.7041 [0.5596, 0.7998]
PIK3CA/mTOR	S4	32 / 32 / 32	0.5166 [0.3730, 0.6602]	0.6641 [0.5263, 0.7910]	0.5166 [0.3730, 0.6523]
PIK3CA/mTOR	S5	32 / 32 / 32	0.5977 [0.4492, 0.7344]	0.7148 [0.5801, 0.8350]	0.5977 [0.4482, 0.7285]
AChE/BChE	S1	27 / 25 / 28	0.6341 [0.4756, 0.7867]	0.6442 [0.4907, 0.7871]	0.6341 [0.4607, 0.7540]
AChE/BChE	S2	27 / 25 / 27	0.5052 [0.3392, 0.6667]	0.4925 [0.3333, 0.6502]	0.4925 [0.3141, 0.5867]
AChE/BChE	S3	27 / 25 / 28	0.6474 [0.4903, 0.7881]	0.4907 [0.3360, 0.6468]	0.4907 [0.3360, 0.6296]
AChE/BChE	S4	27 / 25 / 27	0.5437 [0.3822, 0.6978]	0.3909 [0.2401, 0.5528]	0.3909 [0.2401, 0.5336]
AChE/BChE	S5	27 / 27 / 28	0.6228 [0.4650, 0.7668]	0.4511 [0.2923, 0.6111]	0.4511 [0.2923, 0.5967]
F2/F10	S1	32 / 32 / 32	0.4082 [0.2656, 0.5557]	0.3623 [0.2363, 0.5049]	0.3623 [0.2305, 0.4805]
F2/F10	S2	31 / 32 / 32	0.5383 [0.3911, 0.6774]	0.5675 [0.4194, 0.7067]	0.5383 [0.3821, 0.6300]
F2/F10	S3	32 / 32 / 32	0.6094 [0.4668, 0.7490]	0.6230 [0.4785, 0.7568]	0.6094 [0.4531, 0.7021]
F2/F10	S4	31 / 32 / 32	0.6341 [0.4919, 0.7641]	0.5696 [0.4173, 0.7127]	0.5696 [0.4163, 0.6855]
F2/F10	S5	31 / 32 / 32	0.5625 [0.4133, 0.7107]	0.5554 [0.4062, 0.7026]	0.5554 [0.3921, 0.6633]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.

Table S10 Directional discrimination and paired scoring-workflow comparisons (continued)

A. Method-specific estimates

Target pair	Scheme	n (dual/A/B)	AUROC B [95% CI]	AUROC A [95% CI]	Min. AUROC [95% CI]
PPARG/PPARA	S1	32 / 32 / 32	0.6553 [0.5117, 0.7861]	0.6021 [0.4614, 0.7402]	0.6021 [0.4561, 0.7051]
PPARG/PPARA	S2	30 / 32 / 32	0.6583 [0.5177, 0.7885]	0.2802 [0.1573, 0.4156]	0.2802 [0.1573, 0.4156]
PPARG/PPARA	S3	32 / 32 / 32	0.6455 [0.5029, 0.7744]	0.3105 [0.1748, 0.4492]	0.3105 [0.1748, 0.4492]
PPARG/PPARA	S4	30 / 32 / 32	0.5781 [0.4313, 0.7229]	0.3312 [0.2000, 0.4719]	0.3312 [0.2000, 0.4698]
PPARG/PPARA	S5	32 / 32 / 32	0.5146 [0.3701, 0.6543]	0.4463 [0.3066, 0.5957]	0.4463 [0.3037, 0.5596]
PPARA/PPARD	S1	32 / 32 / 32	0.6787 [0.5430, 0.8047]	0.5107 [0.3652, 0.6543]	0.5107 [0.3652, 0.6475]
PPARA/PPARD	S2	27 / 32 / 32	0.7002 [0.5590, 0.8287]	0.6817 [0.5417, 0.8113]	0.6817 [0.5255, 0.7546]
PPARA/PPARD	S3	32 / 32 / 32	0.7520 [0.6240, 0.8653]	0.6689 [0.5332, 0.7979]	0.6689 [0.5322, 0.7725]
PPARA/PPARD	S4	27 / 32 / 32	0.7581 [0.6296, 0.8738]	0.5625 [0.4144, 0.7118]	0.5625 [0.4144, 0.7025]
PPARA/PPARD	S5	32 / 32 / 32	0.6338 [0.4922, 0.7676]	0.6670 [0.5264, 0.7998]	0.6338 [0.4844, 0.7197]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.

Table S10 Directional discrimination and paired scoring-workflow comparisons

B. Paired differences in minimum AUROC

Target pair	Contrast	n (dual/A/B)	Comparison minimum AUROC	Reference minimum AUROC	Difference (95% CI)
EGFR/HER2	S2-S1	28 / 37 / 32	0.2210	0.2467	-0.0257 [-0.1775, 0.1283]
EGFR/HER2	S3-S1	28 / 37 / 32	0.2634	0.2467	0.0167 [-0.1340, 0.1685]
EGFR/HER2	S4-S2	28 / 37 / 32	0.2812	0.2210	0.0603 [-0.0536, 0.1797]
EGFR/HER2	S5-S1	28 / 37 / 32	0.2455	0.2467	-0.0011 [-0.1629, 0.1596]
JAK1/JAK2	S2-S1	32 / 32 / 32	0.4385	0.5908	-0.1523 [-0.3672, 0.0605]
JAK1/JAK2	S3-S1	32 / 32 / 32	0.4736	0.5908	-0.1172 [-0.2861, 0.0498]
JAK1/JAK2	S4-S2	32 / 32 / 32	0.4443	0.4385	0.0059 [-0.1328, 0.1406]
JAK1/JAK2	S5-S1	32 / 32 / 32	0.5137	0.5908	-0.0771 [-0.2920, 0.1396]
JAK1/TYK2	S2-S1	32 / 32 / 32	0.5596	0.4219	0.1377 [-0.0371, 0.2969]
JAK1/TYK2	S3-S1	32 / 32 / 32	0.4873	0.4219	0.0654 [-0.0791, 0.2109]
JAK1/TYK2	S4-S2	32 / 32 / 32	0.6758	0.5596	0.1162 [-0.0127, 0.2539]
JAK1/TYK2	S5-S1	32 / 32 / 32	0.6689	0.4219	0.2471 [0.0312, 0.4082]
PIK3CA/mTOR	S2-S1	32 / 32 / 32	0.5752	0.5801	-0.0049 [-0.1924, 0.1680]
PIK3CA/mTOR	S3-S1	32 / 32 / 32	0.7041	0.5801	0.1240 [-0.0469, 0.2607]
PIK3CA/mTOR	S4-S2	32 / 32 / 32	0.5166	0.5752	-0.0586 [-0.1660, 0.0547]
PIK3CA/mTOR	S5-S1	32 / 32 / 32	0.5977	0.5801	0.0176 [-0.1807, 0.2178]
AChE/BChE	S2-S1	27 / 25 / 27	0.4925	0.6341	-0.1416 [-0.3676, 0.0489]
AChE/BChE	S3-S1	27 / 25 / 28	0.4907	0.6341	-0.1433 [-0.2608, 0.0172]
AChE/BChE	S4-S2	27 / 25 / 27	0.3909	0.4925	-0.1015 [-0.2215, 0.0923]
AChE/BChE	S5-S1	27 / 25 / 28	0.4511	0.6341	-0.1830 [-0.4026, 0.0715]
F2/F10	S2-S1	31 / 32 / 32	0.5383	0.3740	0.1643 [-0.0141, 0.3054]
F2/F10	S3-S1	32 / 32 / 32	0.6094	0.3623	0.2471 [0.0654, 0.3896]
F2/F10	S4-S2	31 / 32 / 32	0.5696	0.5383	0.0312 [-0.0796, 0.1724]
F2/F10	S5-S1	31 / 32 / 32	0.5554	0.3740	0.1815 [-0.0071, 0.3357]
PPARG/PPARA	S2-S1	30 / 32 / 32	0.2802	0.5891	-0.3089 [-0.4688, -0.1313]
PPARG/PPARA	S3-S1	32 / 32 / 32	0.3105	0.6021	-0.2915 [-0.4380, -0.1104]
PPARG/PPARA	S4-S2	30 / 32 / 32	0.3312	0.2802	0.0510 [-0.0719, 0.1740]
PPARG/PPARA	S5-S1	32 / 32 / 32	0.4463	0.6021	-0.1558 [-0.3457, 0.0459]
PPARA/PPARD	S2-S1	27 / 32 / 32	0.6817	0.5625	0.1192 [-0.0706, 0.2697]
PPARA/PPARD	S3-S1	32 / 32 / 32	0.6689	0.5107	0.1582 [0.0059, 0.2979]
PPARA/PPARD	S4-S2	27 / 32 / 32	0.5625	0.6817	-0.1192 [-0.2616, 0.0868]
PPARA/PPARD	S5-S1	32 / 32 / 32	0.6338	0.5107	0.1230 [-0.0762, 0.2754]

AUROC, target B: dual versus A-only; AUROC, target A: dual versus B-only. Counts: dual/A-only/B-only. S1, Vina; S2, fixed-pose GNINA; S3, fixed-pose RTMScore; S4, maximum saved-pose CNNscore; S5, independent GNINA. Higher scores are favorable. Paired contrasts use their own shared members and need not equal differences between estimates in panel A. CI: 95% class-stratified percentile bootstrap (10,000 resamples; seed 271828). S4-S2: saved-pose reranking. Supporting S4-S1 in Table S9 has no added CI.

Table S11. Exploratory single-target controls

A. Single-target-active versus neither

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)
EGFR/HER2	EGFR	A-only	37	12	0.5766 [0.3761, 0.7680]
EGFR/HER2	HER2	B-only	32	12	0.8672 [0.7109, 0.9818]
JAK1/JAK2	JAK1	A-only	32	14	0.6049 [0.4241, 0.7790]
JAK1/JAK2	JAK2	B-only	32	14	0.5446 [0.3504, 0.7388]
JAK1/TYK2	JAK1	A-only	32	14	0.7455 [0.5558, 0.9085]
JAK1/TYK2	TYK2	B-only	32	14	0.7522 [0.5804, 0.8996]
PIK3CA/mTOR	PIK3CA	A-only	32	14	0.5357 [0.3549, 0.7121]
PIK3CA/mTOR	mTOR	B-only	32	14	0.6183 [0.4576, 0.7768]
AChE/BChE	AChE	A-only	25	15	0.6213 [0.4427, 0.7867]
AChE/BChE	BChE	B-only	28	15	0.6071 [0.4310, 0.7786]
F2/F10	F2	A-only	32	14	0.6339 [0.4621, 0.7879]
F2/F10	F10	B-only	32	14	0.6406 [0.4353, 0.8304]
PPARG/PPARA	PPARG	A-only	32	14	0.6685 [0.4631, 0.8504]
PPARG/PPARA	PPARA	B-only	32	14	0.6384 [0.4330, 0.8281]
PPARA/PPARD	PPARA	A-only	32	14	0.5223 [0.3303, 0.7143]
PPARA/PPARD	PPARD	B-only	32	14	0.7009 [0.5290, 0.8527]

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.

Table S11. Exploratory single-target controls

B. Single-target-active versus opposite single-target-active

Target pair	Score target	Positive class	n positive	n negative	AUROC (95% CI)	
EGFR/HER2	EGFR	A-only	37	32	0.1807 [0.0845, 0.2914]	
EGFR/HER2	HER2	B-only	32	37	0.7939 [0.6748, 0.8970]	
JAK1/JAK2	JAK1	A-only	32	32	0.6250 [0.4844, 0.7588]	
JAK1/JAK2	JAK2	B-only	32	32	0.3755 [0.2417, 0.5186]	
JAK1/TYK2	JAK1	A-only	32	32	0.3125 [0.1865, 0.4512]	
JAK1/TYK2	TYK2	B-only	32	32	0.6777 [0.5410, 0.8057]	
PIK3CA/mTOR	PIK3CA	A-only	32	32	0.4268 [0.2900, 0.5713]	
PIK3CA/mTOR	mTOR	B-only	32	32	0.6348 [0.4941, 0.7686]	
AChE/BChE	AChE	A-only	25	28	0.7000 [0.5500, 0.8286]	
AChE/BChE	BChE	B-only	28	25	0.5293 [0.3629, 0.6900]	
F2/F10	F2	A-only	32	32	0.4619 [0.3213, 0.6084]	
F2/F10	F10	B-only	32	32	0.5039 [0.3584, 0.6445]	
PPARG/PPARA	PPARG	A-only	32	32	0.5044 [0.3584, 0.6475]	
PPARG/PPARA	PPARA	B-only	32	32	0.5083 [0.3623, 0.6523]	
PPARA/PPARD	PPARA	A-only	32	32	0.5542 [0.4102, 0.6958]	
PPARA/PPARD	PPARD	B-only	32	32	0.7144 [0.5801, 0.8345]	

S1: negative Vina docking energy, with higher values favorable (search seed 42). The positive class is the single-target class active at the scored target. Neither ligands are negative in panel A; the opposite single-target class is negative in panel B. Dual ligands are excluded. Counts include eligible observations with a valid score at that target. CI: 95% class-stratified percentile bootstrap, 10,000 resamples, seed 271828; linear 2.5th and 97.5th percentiles. Positive resamples are shared between the two comparisons. Score ties count as 0.5. Intervals are exploratory and unadjusted for multiplicity.